

SACUIENI, Romania

WMO: 150420

Lat: 47.3442N

Lon: 22.0944E

Elev: 126

StdP: 99.83

Time Zone: 2.00 (EUE)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.4	-8.5	-14.4	1.1	-9.9	-11.3	1.5	-7.1	10.4	5.4	9.1	4.7	1.3	50	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.2	32.6	21.8	31.0	21.3	29.5	20.6	23.2	30.4	22.2	28.9	21.4	27.8	2.2	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.0	15.9	26.6	20.0	14.9	25.6	19.1	14.1	24.5	69.5	30.0	65.7	29.3	62.7	27.8	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.6	7.5	6.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	-1.0	2.5	6.5	11.8	16.9	20.2	21.2	21.7	16.7	11.4	5.5	1.2
	DBStd	8.92	4.61	4.76	4.59	4.64	3.58	3.34	3.17	3.52	3.89	4.50	5.11	4.74
	HDD10.0	1188	342	212	126	36	2	0	0	0	1	38	153	277
	HDD18.3	2934	600	444	368	201	69	19	9	9	78	217	386	532
	CDD10.0	1642	0	1	16	88	215	306	348	364	201	82	18	3
	CDD18.3	347	0	0	0	3	24	74	99	115	28	2	0	0
	CDH23.3	3330	0	0	1	31	206	664	919	1206	277	27	1	0
	CDH26.7	1155	0	0	0	2	31	213	316	510	83	0	0	0
(14) Wind	WSAvg	2.3	2.1	2.7	2.7	2.8	2.6	2.0	1.9	1.7	1.8	2.0	2.3	2.5
(15) Precipitation	PrecAvg	590	31	30	33	44	57	79	71	49	56	49	48	48
	PrecMax	854	63	93	140	71	128	153	127	141	163	109	84	109
	PrecMin	383	11	2	7	6	11	13	11	3	9	3	0	3
	PrecStd	129	15	21	32	18	29	41	33	35	36	31	22	27
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.3	14.7	21.8	26.1	29.3	33.0	33.7	35.7	31.9	25.7	19.5	15.4
		MCWB	8.5	9.9	13.1	16.3	18.7	22.8	22.1	23.1	19.8	16.3	13.4	10.8
	2%	DB	8.9	12.3	18.3	23.8	27.5	31.2	31.8	33.3	28.8	23.9	17.0	12.1
		MCWB	6.9	8.9	11.5	15.1	18.3	22.0	21.9	21.6	18.6	15.7	12.5	8.9
	5%	DB	7.1	10.9	15.5	21.4	25.7	29.3	30.0	31.4	26.7	21.6	14.9	9.7
		MCWB	5.4	7.9	10.1	13.9	17.9	21.1	21.3	21.2	18.1	14.8	11.5	7.3
	10%	DB	5.3	9.2	13.2	19.5	24.0	27.2	28.3	29.8	24.0	19.1	12.7	7.1
		MCWB	4.0	6.7	8.6	12.7	16.7	20.1	20.5	20.4	17.3	13.6	10.3	5.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.7	11.0	14.0	17.3	20.2	24.7	24.6	24.1	20.6	17.9	13.9	11.2
		MCDB	10.9	14.1	20.7	24.2	26.4	31.4	31.8	33.6	28.1	21.7	18.0	15.0
	2%	WB	7.4	9.3	12.1	15.9	19.2	23.0	23.0	23.0	19.5	16.2	12.9	9.3
		MCDB	9.2	11.9	17.2	22.5	25.6	29.4	28.8	31.3	26.7	22.8	16.7	11.8
	5%	WB	5.5	8.2	10.4	14.6	18.3	21.8	22.0	22.0	18.7	15.3	11.9	7.4
		MCDB	6.8	10.6	14.8	20.7	24.0	27.7	28.0	29.9	25.2	20.9	14.7	9.6
	10%	WB	3.9	7.0	9.1	13.4	17.5	20.8	21.2	21.1	17.7	14.0	10.8	5.8
		MCDB	5.0	8.9	12.6	18.7	22.6	26.3	27.0	28.4	23.2	18.6	13.0	7.2
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	6.5	8.8	10.6	10.9	10.8	11.4	12.2	10.6	10.6	7.0	4.8
		MCDBR	6.3	8.7	11.9	13.2	12.9	13.2	13.7	14.7	14.8	14.4	9.2	6.8
		MCWBR	4.4	6.0	7.0	6.7	5.9	5.8	5.6	5.4	6.2	7.7	5.2	4.9
	5% WB	MCDBR	5.7	8.4	11.4	12.3	12.0	12.3	12.0	13.8	13.6	13.8	8.4	6.4
		MCWBR	4.3	5.9	6.9	6.4	5.8	5.8	5.3	5.5	5.7	7.3	4.9	4.7
(40) Clear-Sky Solar Irradiance	taub		0.332	0.358	0.397	0.450	0.434	0.434	0.453	0.451	0.417	0.385	0.358	0.333
	taud		2.314	2.272	2.201	2.078	2.193	2.224	2.180	2.181	2.242	2.309	2.331	2.344
	Ebn at Noon		731	790	812	801	833	833	810	793	784	749	692	683
	Edh at Noon		76	98	122	151	140	137	141	135	116	93	75	67
(44) All-Sky Solar Radiation	RadAvg		1.04	1.88	3.14	4.51	5.51	5.99	5.87	5.37	3.74	2.43	1.33	0.79
	RadStd		0.14	0.29	0.44	0.45	0.54	0.40	0.51	0.48	0.43	0.29	0.16	0.12

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (53 neighbors)	+0.69	N/A	N/A	+1.04	N/A	N/A	-127	-194	+113	+57

Nomenclature: See separate page